

Angelina Caggiano, EIT

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EDUCATION

ANTICIPATED MAY 2026	University of Massachusetts, Amherst <i>Doctor of Philosophy in Civil Engineering (Transportation)</i> Program: Transportation Engineering Chair: Dr. Michael Knodler Research interests: Safety, pedestrians, human factors, commercial vehicles, data analytics
	University of Massachusetts, Amherst <i>Master of Science in Civil Engineering, 3.92 GPA</i> ‘An Evaluation of Driver Comprehension & Behavior at Pedestrian Hybrid Beacon (PHB) Intervals’ Chair: Dr. Michael Knodler
	University of Massachusetts, Amherst <i>Bachelor of Science in Civil Engineering</i> Cum Laude, 3.71 GPA Spring 2020: Attended <i>SKEMA Business School, Sophia Antipolis France</i>

RESEARCH EXPERIENCE

University of Massachusetts Amherst

2022– PRESENT	Graduate Research Assistant Leads real world and laboratory-based research in traffic operations and safety.
	RIDE REU Graduate Mentor <i>Research on Inclusivity and Driving Equity (RIDE) NSF-funded REU Program.</i> Project titles: Pedestrian Behavior at Pedestrian Hybrid Beacons, Improving Safe Driving Media Campaigns Using Psychology and Demographic Targeting, Public Transit Safety Among Communities in Massachusetts.
SUMMER 2023 & 2024	
2020– 2022	Undergraduate Research Assistant Project titles: Mount Ida Campus Shuttle Program, MassDOT Intersection Inventory, Fatal and Serious Injury Crash Differences by Gender

PROFESSIONAL EXPERIENCE

Kimley-Horn

JUNE 2025 – AUGUST 2025	<i>Research Analyst Intern</i> Contributed to several NCHRP deliverables including final deliverable “Research Implementation Toolkit: Strategies for Overcoming Barriers” for NCHRP 20-44(49) and TFPE 08 proposal “Evaluation of the FHWA Highway Safety Information Systems” (\$300,000 grant).
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Vanasse Hangen Brustlin (VHB) Inc.

JANUARY 2023– SEPTEMBER 2023	<i>Highway Safety Intern</i> Analyzed crash data to prepare reports, sketches, and memorandums related to complete street safety analysis.
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Arcadis IBI Group

JUNE 2022– DECEMBER 2022	<i>Intelligent Transportation Systems Intern</i> Analyzed and researched DOT performance measures, travel time data, unmanned aircraft systems, systems engineering and requirements documentation, and traffic incident management strategies.
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Massachusetts Department of Transportation

JUNE 2021– AUGUST 2021	<i>Highway Engineering Intern</i> Assisted with inspection of roadway construction, conducted field measurements and field reports.
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TEACHING EXPERIENCE

FALL 2023 & 2024	First Year Seminar Instructor: <i>Transportation Innovation & Smart Cities</i> Independently designed and taught an original course, overseeing all content development and instruction. Held full responsibility for grading and evaluation.
SUMMER 2023	Project Leader: <i>Summer Engineering Institute</i> Designed and led two-week research projects for high school students focused on crowdsourcing application design and machine learning image detection, emphasizing traffic safety applications.
FALL 2022	Teaching Assistant: <i>Traffic Engineering (CEE411/511)</i> Supported course instruction through leading lab sessions, grading assignments, and providing student guidance during office hours.

FELLOWSHIPS & AWARDS

2025	New England University Transportation Center Student of the Year
2025	Fall Civil Engineering Department Travel Grant (\$300)
2025	WTS-Boston Mary Jane O'Meara Graduate Scholarship (\$4,000)
2025	AAA Safe Mobility Conference Student Poster Competition, 2 nd Place (\$800)
2025	Spring Civil Engineering Department Travel Grant (\$368.54)
2025	Dwight David Eisenhower Transportation Graduate Fellowship (\$10,000)
2024	Southeastern New England ITE Graduate Scholarship (\$1,000)
2024, 2023	Traffic Safety Scholar for Lifesavers Conference on Roadway Safety (\$1,000)
2024	SaferSIM University Transportation Center Student of the Year
2024	ATSSA Traffic Control Device Student Competition 1 st Place Award (\$1,500)
2023	Fall Civil Engineering Department Travel Grant (\$210)
2023	WTS-Boston Ann Hershfang Graduate Scholarship (\$4,000)
2022	Berthiaume Center for Entrepreneurship Technology Challenge 2 nd Place Award (\$8,000)
2022	MA ITE Thomas Desjardin Scholarship (\$2,000)
2018-2022	Jack F. Welch Scholarship (Undergraduate, \$15,000/year)

PUBLICATIONS

(* corresponding author)

Journal Articles

1. **Caggiano, A.***, Christofa, E., Tainter, F., & Knodler, M. (2025). *Field Study of Driver Behavior by Interval at Pedestrian Hybrid Beacons*. Transportation Research Record: Journal of the Transportation Research Board, 0(0). [[Link](#)]
2. **Caggiano, A.***, Pamarthi, J., Zafian, T. M., Deng, M. Y., Johnson, K., Tainter, F., & Knodler, M. (2025). *An evaluation of driver comprehension of the Pedestrian Hybrid Beacon*. Journal of Safety Research, 92, 181–191. [[Link](#)]

Manuscripts Under Review

1. **Caggiano, A.***, Gazzillo, J., Roney-Yeager, B., Fitzpatrick, C., Pearlman, J., Christofa, E., Riessman, R., & Knodler, M. (Under Review). *Predictors of Risky Driving Behaviors in Commercial Motor Vehicle Drivers: A Survey-Based Analysis*.
2. **Caggiano, A.***, Long, S., Oke, T., Tainter, F., & Knodler, M. (Under Review). *Pushing the Button or Pushing the Gap? An Analysis of Pedestrian Crossing Behavior at Pedestrian Hybrid Beacons*

Manuscripts in Preparation

1. **Caggiano, A.***, Shoub, K., & Knodler, M. (In Preparation). Scooters, E-Bikes, and Motorized Bikes, Oh My! A Review of Micromobility Device Definitions & Classifications in the United States.

Reports & Projects

1. Johnson, B. (PI), Nguyen, R., D'Sa, K., **Caggiano, A.**, Ivey, S. Dontah, A., & Hammit, D. (Accepted). NHCRP 20-44(49) Final Report: Research Implementation - Strategies for Overcoming Barriers.

RESEARCH GRANTS

1. Johnson-Hammit, B. (PI), Ivey, S., & **Caggiano, A.** TFPE 08 Evaluation of the FHWA Highway Safety Information Systems, TRB/FHWA Program Evaluation, September 2025. \$300,000.
2. **Caggiano, A.** Dwight David Eisenhower Transportation Fellowship, U.S. D.O.T. Grant Agreement 693JJ32545021. 2024–2025. \$10,000.

Grants Under Review

1. Fitzpatrick, C., Knodler, M., Gazzillo, J., & **Caggiano, A.** (Under Review). *Beyond the Shoulder: A Comprehensive Study of the Roadside Service Industry and Safety Culture*. AAA Foundation for Traffic Safety Proposal.

PRESENTATIONS

Conference & Annual Meeting Presentations Accepted by Abstract (*corresponding author(s) & presenter(s))

1. **Caggiano, A.***, Pamarthi, J., Zafian, T., Tainter, F., & Knodler, K. (2025) *Advancing the Understanding of Driver and Pedestrian Behaviors at Pedestrian Hybrid Beacons: Past, Present, and Future Research*. MassDOT Innovation Conference, May 7, 2025, Worcester, MA. (Poster Presentation).
2. **Caggiano, A.***, Pamarthi, J., Zafian, T., Tainter, F., & Knodler, K. (2025) *Advancing the Understanding of Driver and Pedestrian Behaviors at Pedestrian Hybrid Beacons: Past, Present, and Future Research*. AAA Safe Mobility Conference, Madison, WI. (Poster Presentation).
3. **Caggiano, A.***, Christofa, E., Tainter, F., & Knodler, M. (2025) *A Field Study of Driver Behavior by Interval at Pedestrian Hybrid Beacons*. Transportation Research Board 104th Annual Meeting, January 2025, Washington, DC. (Poster Presentation)
4. **Caggiano, A.***, Pamarthi, J., Tainter, F. & Knodler, M. (2024). *Driver Behavior by Interval at Pedestrian Hybrid Beacon Intervals*. Future of Transportation Summit, July 13-15, Washington, D.C. (Poster Presentation).
5. **Caggiano, A.***, Pamarthi, J., Tainter, F. & Knodler, M. (2024). *Driver Behavior by Interval at Pedestrian Hybrid Beacons*. Institute of Transportation Engineers Annual Meeting, July 21, Philadelphia, PA. (Poster Presentation).
6. **Caggiano, A.***, Pamarthi, J., Tainter, F. & Knodler, M. *Driver Behavior by Interval at Pedestrian Hybrid Beacons*. MassDOT Innovation Conference, April 30, 2024, Worcester, MA. (Podium Presentation).
7. **Caggiano, A.***, Pamarthi, J., Zafian, T., Deng, M., Johnson, K., Tainter, F. & Knodler, M. (2024). *An Evaluation of Driver Comprehension of the Pedestrian Hybrid Beacon*. Transportation Research Board 103rd Annual Meeting, January 7-11, Washington, DC. (Poster Presentation).

A. Caggiano

8. Dunn, M.*, **Caggiano, A.**, Porter, R., Kersavage, K., Lim, I., Dadvar, S., & Dimaiuta, M. (2024). *Complete Streets Treatment Combinations and Safety Analysis Needs Assessment*. Transportation Research Board 103rd Annual Meeting, January 7-11, Washington, DC. (Poster Presentation).
9. **Caggiano, A.***, Pamarthi, J., Zafian, T., Deng, M., Johnson, K., Tainter, F. & Knodler, M. (2023). *Evaluation of Driver Comprehension of the Pedestrian Hybrid Beacon*. ITE International Meeting, August 12-16, Portland, OR. (Poster Presentation).
10. **Caggiano, A.***, Knodler, M., & Tainter, F. (2023). *Evaluation of Driver and Pedestrian Comprehension of the Pedestrian Hybrid Beacon*. Northeastern ITE District Meeting, May 10-12, Burlington, VT. (Podium Presentation).
11. **Caggiano, A.***, Knodler, M., & Tainter, F. (2023) *Evaluation of Driver and Pedestrian Comprehension of the Pedestrian Hybrid Beacon*. MassDOT Innovation Conference, May 2-3, Worcester, MA. (Podium Presentation).
12. **Caggiano, A.***, Knodler, M., & Tainter, F. (2023) *Evaluation of Driver and Pedestrian Comprehension of the Pedestrian Hybrid Beacon*. SaferSIM Research Symposium, February 18-20, Mayaguez, PR. (Poster Presentation).

Other Presentations

3. **Caggiano, A.***. (2024) “*Women Leaders: Conversations On Leadership Development*”. Women in ITE, September 10th, virtual. (Panelist).
4. **Caggiano, A.*** (2024). *Ascending Passive Detection to Light the Way of Vulnerable Road Users (GlowSafely)*. American Traffic Safety Services Association National Convention, February 4, San Diego, CA. (Podium Presentation).
5. **Caggiano, A.*** (2024). *Ascending Passive Detection to Light the Way of Vulnerable Road Users (GlowSafely)*. Transportation Research Board 103rd Annual Meeting, January 7-11, Washington, DC. (Podium Presentation).
6. **Caggiano, A.*** (2023). *The Power of Broadening Participation and Diversity Among Student Inventors; Building Pathways to Invention*, October 19, Amherst, MA. (Panelist).

SERVICE

Professional Membership & Involvement

2025– PRESENT	Member, TRB Standing Committee on Traffic Control Devices and Work Zone Management (ACF12)
2024– PRESENT	Member, American Society of Civil Engineers
2023– PRESENT	Reviewer, Journal of Transportation Engineering
	Student Affiliate, Transportation Research Board
	Paper Reviewer, Transportation Research Board Annual Meeting
	Friend of Standing Committee on Pedestrians
2021– PRESENT	Friend of Standing Committee on Human Factors of Infrastructure Design and Operations
	Friend of Standing Committee on Traffic Control Devices
	Friend of Standing Committee on Research Innovation Implementation Management
	Friend of Standing Committee on Women and Gender in Transportation
2021– PRESENT	Member, Institute of Transportation Engineers
2020– PRESENT	Member, Women Transportation Seminar

Conference Service & Leadership

2025	Lead Organizer for the Women in CEE Leadership Summit, April 11, UMass Amherst. Awarded Women For UMass Grant (\$5,000)
2024 & 2025	Reviewer, New England UTC Request for Proposals
2024	ITE International Annual Meeting Abstract Reviewer

A. Caggiano

2023 & 2025	Student Lead, MassDOT Moving Together Conference, Boston, MA.
2023– 2025	Founding Member, Civil Engineering Graduate Student Organization
2023– 2025	Secretary, Civil Engineering Graduate Student Organization
2023	Student Lead, MassDOT Transportation Innovation Conference, Worcester, MA.
2023	Local Arrangements Committee, Northeastern District ITE
2023	Session Moderator, “ <i>Retaining a Diverse Team through Diverse Leadership</i> ”, May 12, ITE
2023	Northeastern District Annual Meeting, Burlington, VT.
2023	Organizer of the student poster session at Northeastern ITE district meeting
2023	Abstract Reviewer for the 7 th International Conference on Women and Gender in Transportation
2022– 2024	UMass Student Chapter President, Institute of Transportation Engineers
2022, 2023 & 2024	CEE DEI Winter Formal, Awarded CEE Diversity, Equity, and Inclusivity Grant (\$3,000)
2021– 2022	Student Chapter Secretary, Institute of Transportation Engineers
2020– 2021	Student Chapter Secretary, Women Transportation Seminar

PUBLIC MEDIA APPEARANCES

2025	“Crosswalk Confusion? How pedestrian hybrid beacons work,” by Catherine Parrotta, Boston 25 News. [Link]
2025	“Drivers can't figure out new pedestrian crosswalks,” by Mike Deehan, Axios Boston. [Link]
2025	“What's a pedestrian hybrid beacon? The traffic light Mass. drivers keep messing up,” by Nik DeCosta-Klipa & Hanna Ali, WBUR. [Link]
2025	“‘Confusing’: What readers think of hybrid traffic lights,” by Frances Klemm, Boston.com 30. [Link]
2025	“UMass study finds drivers often confused by crosswalk beacons,” by Scott Merzbach, Daily Hampshire Gazette. [Link]
2025	“‘Confusion at the crosswalk’: Drivers, pedestrians unsure about crossing beacons,” by Namu Sampath, MassLive. [Link]
2025	“New kind of traffic signal is confusing Massachusetts drivers as 25% run red light, study finds,” by Neal Riley & Courtney Cole, WBZ News. [Link]
2025	“Navigating the Unknown: UMass Amherst Researchers Uncover Driver Confusion Surrounding Pedestrian Hybrid Beacons in Massachusetts,” by Bioengineer.org. [Link]
2025	“Study: Nearly 25% of Massachusetts drivers run red lights at pedestrian hybrid beacons,” by Ashley Shook, WWLP. [Link]
2025	“Crosswalk Confusion: UMass Amherst Researchers Find Massachusetts Drivers Flummoxed by Pedestrian Hybrid Beacons,” by Julia Westbrook, University of Massachusetts Amherst, News. [Link]
2025	“CEE’s Angelina Caggiano Publishes Paper on “Pedestrian Hybrid Beacons” in Journal of Safety Research,” Riccio College of Engineering, News. [Link]
2024	“Crossroads Of Transportation Engineering And Human Interactions. UMass Amherst PhD Student Angelina Caggiano is Investigating Roadway Safety,” UMass Graduate School, Stories. [Link]
2023	“PhD Student, Angelina Caggiano at the 2023 ITE Annual Meeting & Exhibition,” YouTube, Applied Information Inc. [Link]

ADVISING

Undergraduate Research Assistants

2024– PRESENT	Sarah Long (<i>Civil Engineering undergraduate student</i>)
2024– 2025	Julian Bianci (<i>Civil Engineering undergraduate student</i>)
2024– 2025	Tori Farrell (<i>Civil Engineering undergraduate student</i>)
2024– 2025	Brooke DeJesus (<i>Civil Engineering undergraduate student</i>)
2024– 2025	Sophia Lucarelli (<i>Civil Engineering undergraduate student</i>)
2024– 2025	Catelyn Yam (<i>Computer Engineering undergraduate student</i>)
2024– PRESENT	Cynthia Elsie Kaleebu (<i>Civil Engineering undergraduate student</i>)
2024	Otto Weigel (<i>Physics and Mathematics undergraduate student</i>)
2024	Aaron Rodriguez Alvarez (<i>Psychology undergraduate student</i>)
2024	Meghan Hausser (<i>Psychology undergraduate student</i>)
2023	Catalina Macedo Giang (<i>Geography undergraduate student</i>)
2021– PRESENT	Grace Letendre (<i>Civil Engineering undergraduate student</i>)